SL Assignment 1

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Introduction

This reporrt analyses and models median house proces across a range of features inlcluding taxes, property and crime rates. The median house value is regressed against these features using several regression models, namely staight linear regression using all features, backwise step regression, ridge regression and lasso regression. The out of sample mean sqaured errrs are assesed across the the results of these models. The inclusion of an interaction variables, chas_rm, is added to the best performing model*

Exploratory Data Analysis

For the sake of data and analysis completeness, EDA is conducted on the entire boston dataset.

Data Attributes

Table 1: boston data attributes

row_count	column_count
506	13

Table 2: Boston Data Data Summary

feature	min	q25	mean	meadian	q75	max	sd	datatype
crim	0.00632	0.082045	3.6135236	0.25651	3.677083	88.9762	8.601744	numeric
zn	0.00000	0.000000	11.3636364	0.00000	12.500000	100.0000	23.322521	numeric
indus	0.46000	5.190000	11.1367787	9.69000	18.100000	27.7400	6.860029	numeric
chas	0.00000	0.000000	0.0691700	0.00000	0.000000	1.0000	0.244949	integer
nox	0.38500	0.449000	0.5546951	0.53800	0.624000	0.8710	0.100000	numeric

rm	3.56100	5.885500	6.2846344	6.20850	6.623500	8.7800	0.700000	numeric
age	2.90000	45.025000	68.5749012	77.50000	94.075000	100.0000	28.148890	numeric
dis	1.12960	2.100175	3.7950427	3.20745	5.188425	12.1265	2.104757	numeric
rad	1.00000	4.000000	9.5494071	5.00000	24.000000	24.0000	8.707468	integer
tax	187.00000	279.000000	408.2371542	330.00000	666.000000	711.0000	168.537118	integer
ptratio	12.60000	17.400000	18.4555336	19.05000	20.200000	22.0000	2.165641	numeric
lstat	1.73000	6.950000	12.6530632	11.36000	16.955000	37.9700	7.140728	numeric
medv	5.00000	17.025000	22.5328063	21.20000	25.000000	50.0000	9.197282	numeric

The complete data set contains 506 observations across 13 columns. The data is complete, with no null values, and all columns are registered as numeric. **Tax (28404), age (792), and zn (543)** display the greatest variance across the features, whilst **nox (0.01) and rm (0.49)** show the least variance across their respective values.

Median House Value Distributions

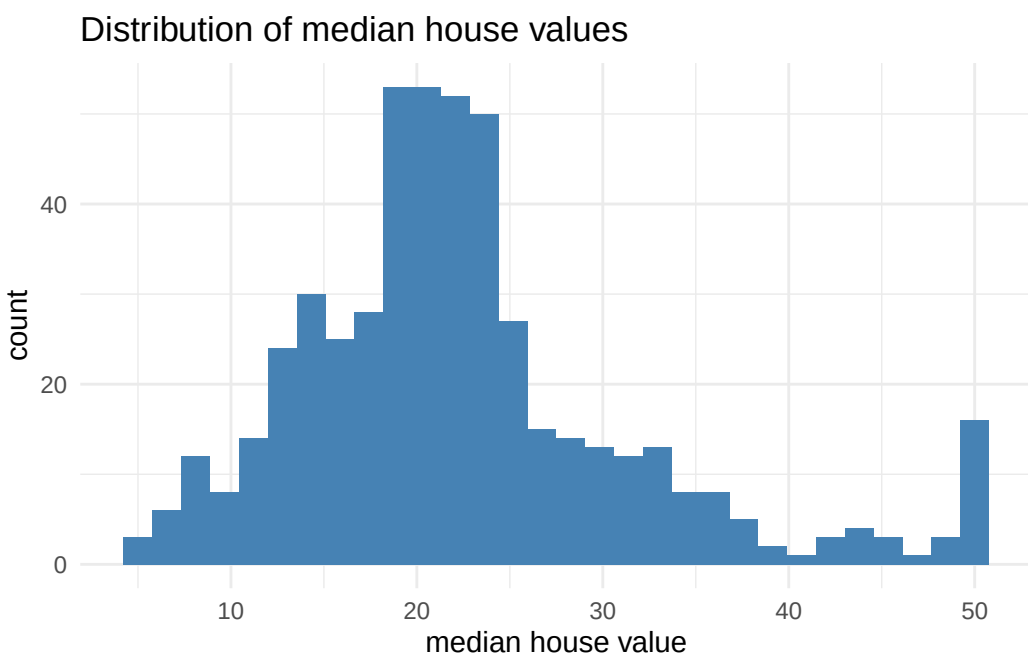


Figure 1: Median House Value Distribution

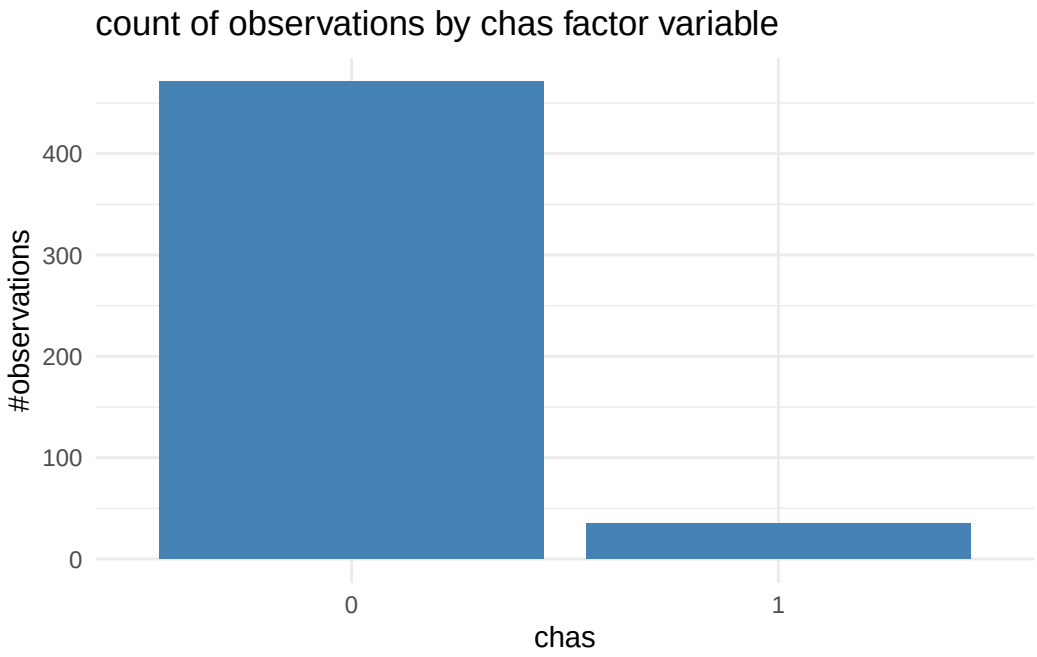


Figure 2: Observation Count Chas Factor

The data data is imbalanced, with more observations for the chas class=0.

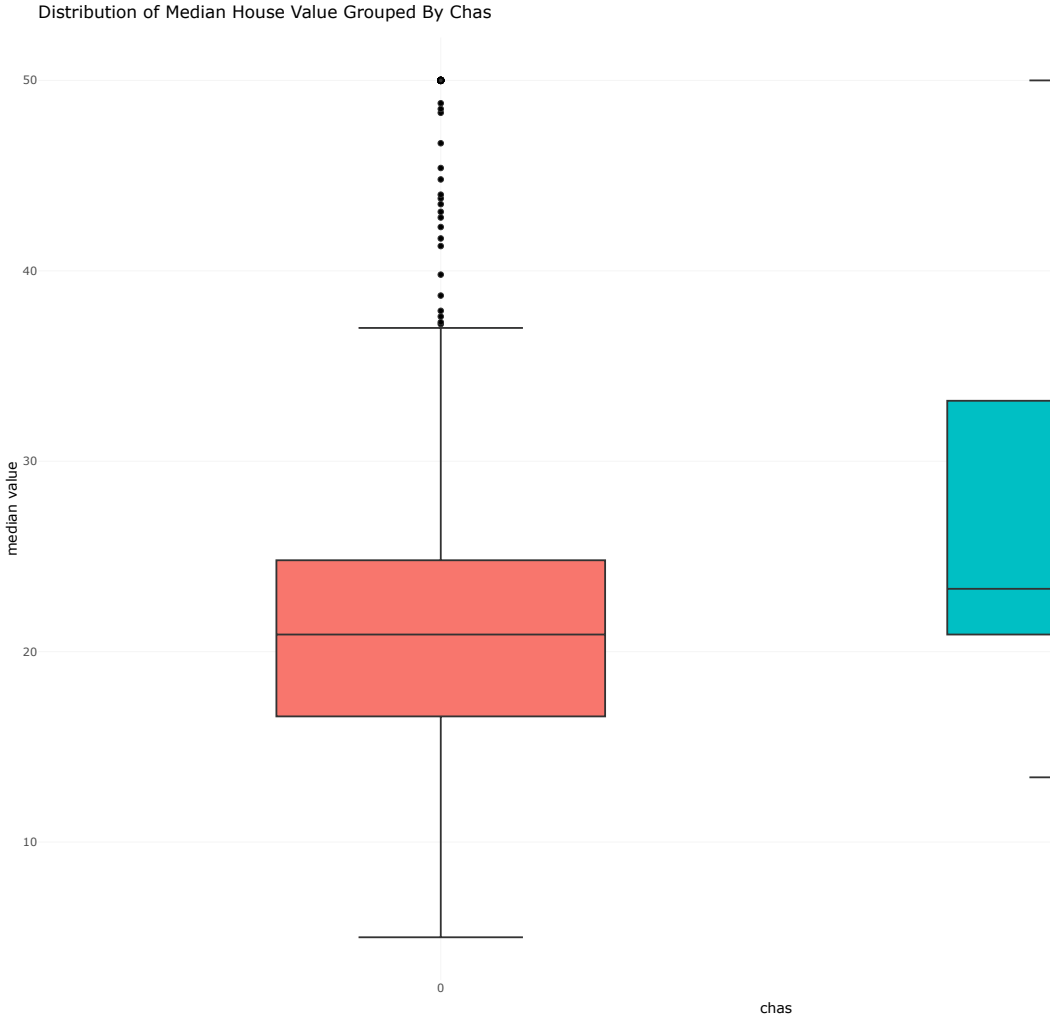


Figure 3: Median House Value Distribution,Grouped by Chas

The median house value for observations with chas=0 is slightly higher at 23.3K, and the observations in this group skew towards higher values, in contrast to the distribution of the chas=1 observations, which is relatively more balanced around the median. Despite the difference in distributions, the two box plots show some overlap, which could indicate that the difference in means of these two groups may not be statistically significant.

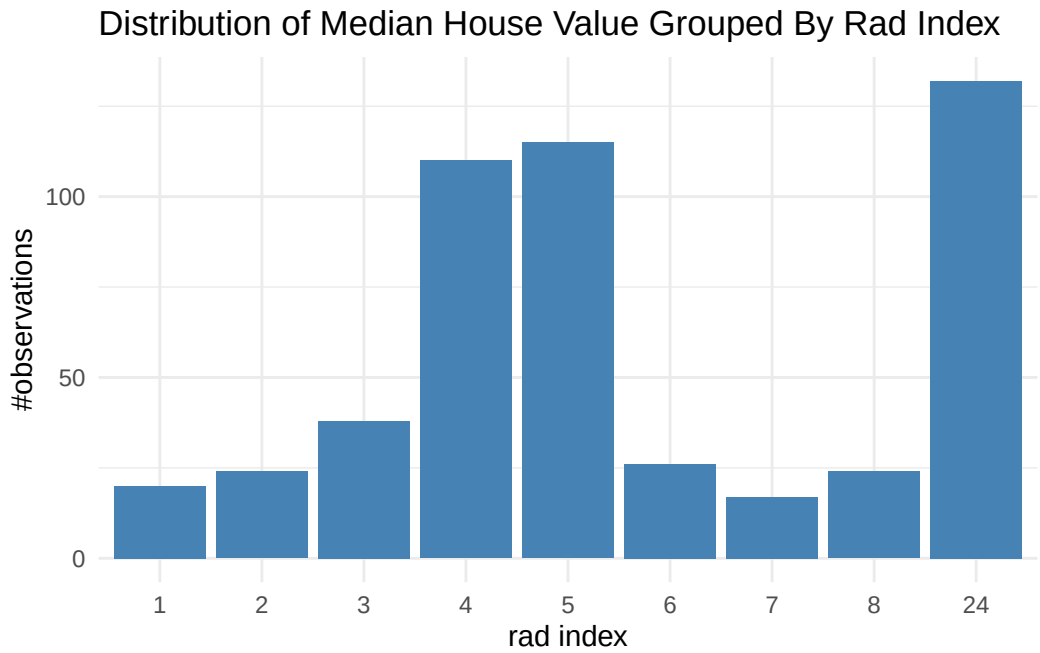
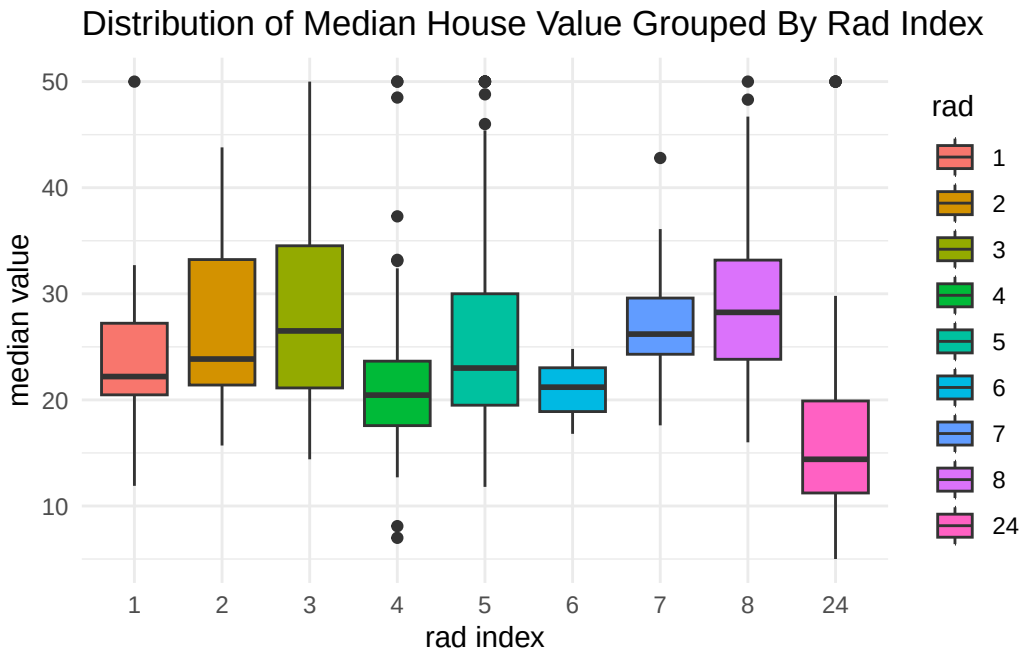


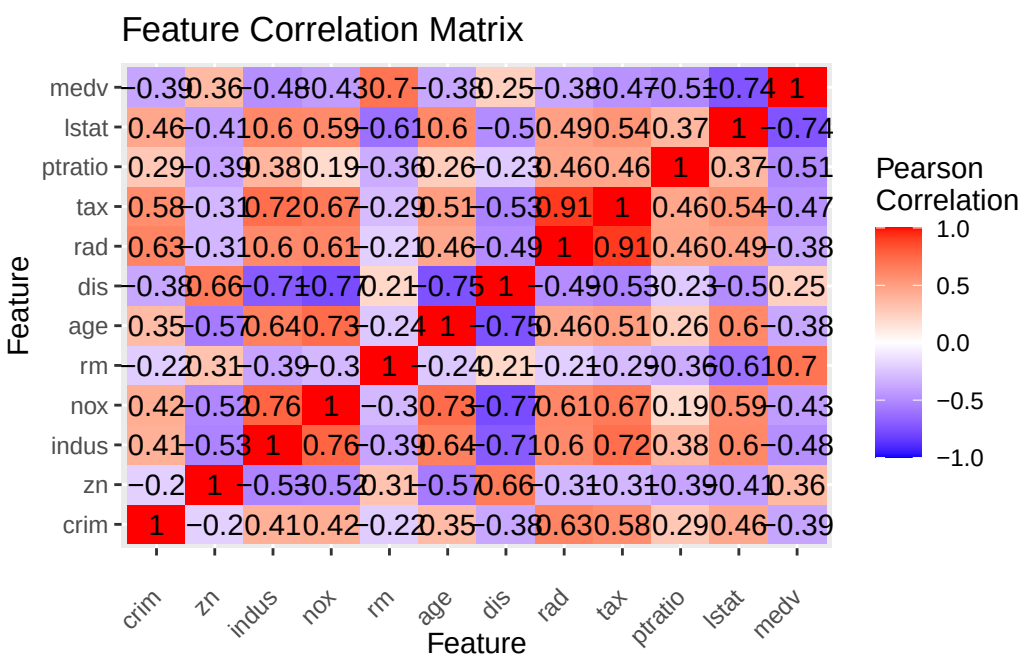
Figure 4: Observation Count Per Radial Accessibility Index

Most observations are contained with the rad values= 4,5 and 24



Grouping house value distributions by the RAD index value shows that median house values do not consistently increase or decrease with changes in the rad index. Observations with a rad value of 24 display the lowest house values among the index groups.

Feature Correlations



Features, **rm and lstat** display the largest correlation, in absolute value, with medv. the rad and tax variables are most most highly correlated variables amongst features- signaling a strong possibility of multicollinearity *this could what is causing our variance to blow up-talk about the implications

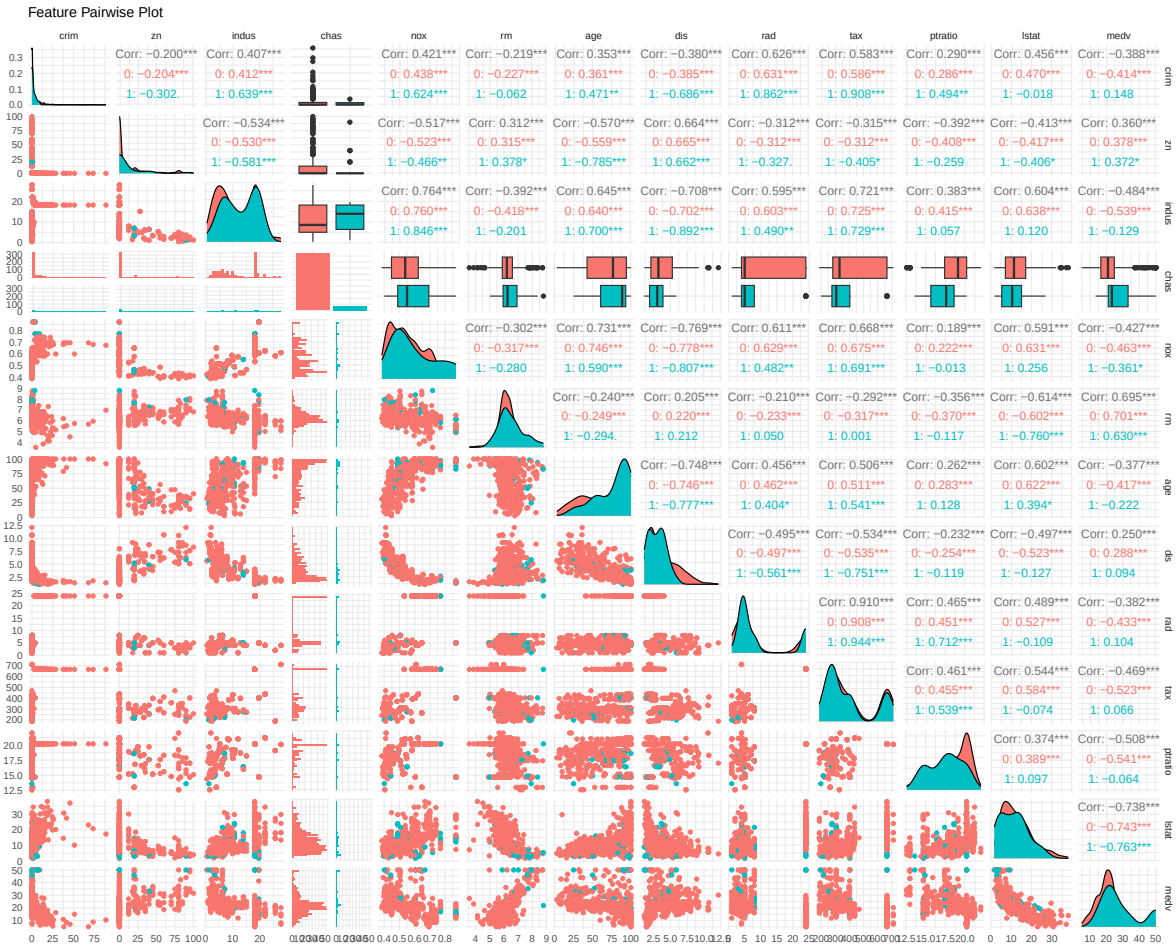


Figure 5: Feature Pairplot

Analyzing the significance of the the correlations, the correlations between the response variable, medv, and all the other numeric features look to be significant at 5%.

Model Building

```
Baseline Model:All Feature Regression Summary
```

	medv
crim	-0.13***
	(0.04)
zn	0.06***
	(0.02)
indus	0.03
	(0.08)
chas1	1.77*
	(0.96)
nox	-20.60***
	(4.41)
rm	3.75***
	(0.49)
age	-0.01
	(0.02)
dis	-1.65***
	(0.23)
rad	0.29***
	(0.08)
tax	-0.01***
	(0.005)
ptratio	-1.01***
	(0.15)
lstat	-0.48***
	(0.06)
Constant	44.45***
	(5.65)
N	400

```
R2                                  0.74
Adjusted R2                         0.73
Residual Std. Error            4.84 (df = 387)
F Statistic                90.99*** (df = 12; 387)
=========================================================
Notes:              ***Significant at the 1 percent level.
                     **Significant at the 5 percent level.
                      *Significant at the 10 percent level.
```

Model Performance:

- The model shows a relatively good fit to the training data, with ~74% of the variation in the data, explained by the model
- 10 out of the 12 feature coefficients (excluding the intercept) are statistically different from 0 at at lest the 10% level, with 9 of these significant at the 1% level
- Nitrogen Oxide (nox) levelsshow the biggest marginal change , in absolute value, to median house values, medv, with an estimated negative change of -20.6K in estimated average houspe values per one unit increase in nox levels.
- The F statistic is also significant at the 1% level, indicating that the coefficients of the model are jointly statistically different from 0.

Prediction of test data set

Table 3: Baseline Test MSE

baseline.mse
16.64

The out-of-sample (OOO) MSE for the baseline model is at 16.64K, that is, on average the baseline model's home values predictions are expected to deviate from the actual by this value, which is greater that the standard deviation of median house values of ~9.2.

Model Improvement

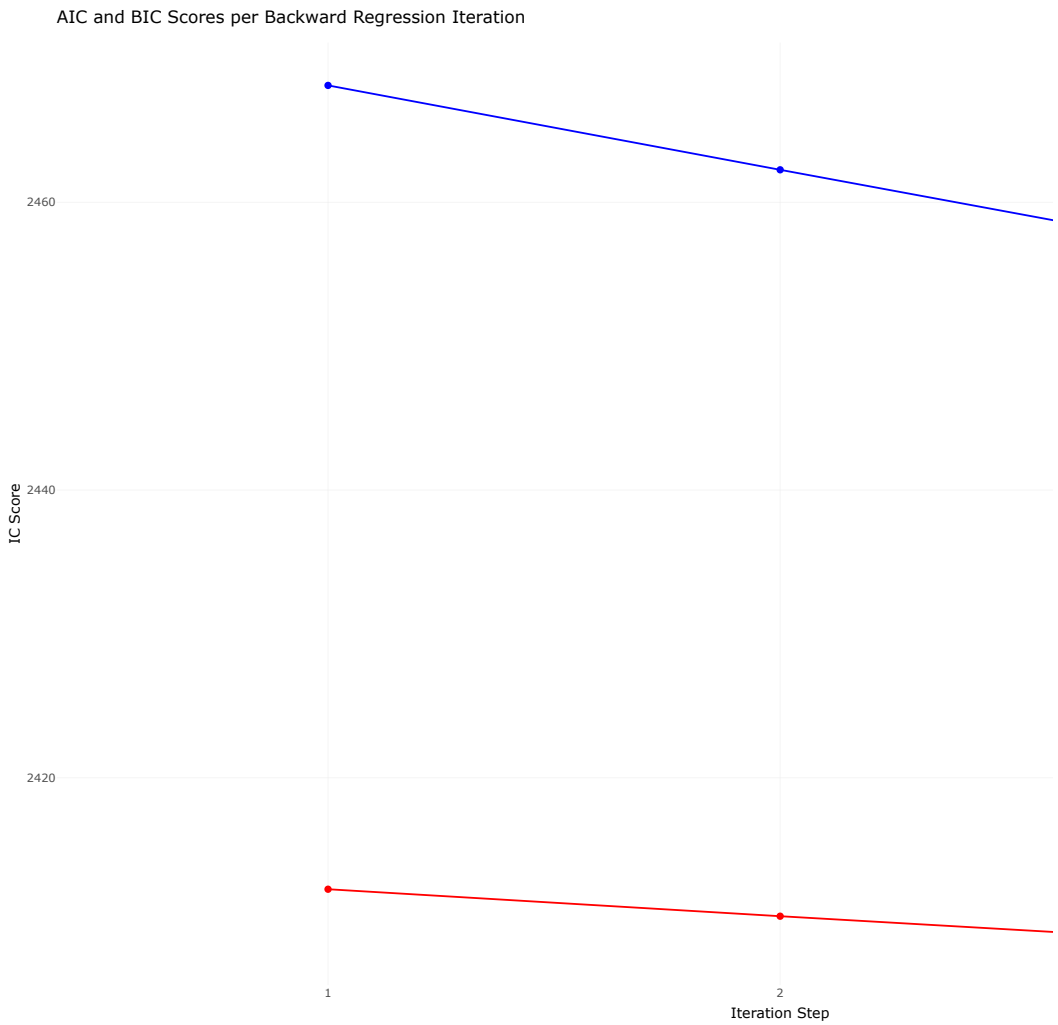
Backwards Step wise Variable Selection

Method:

1. Start off with the baseline model: regress response variable against all features

2. Iteratively remove one feature, retrain the model and store both the AIC and BIC scores for each retrained model.

ref website:https://alain-vandormael.netlify.app/post/varselect/#:~:text=Stepwise%20regression,next%20deletion%20increases%20the%20AIC.


AIC and BIC Scores per Backward Regression Iteration
2460
2440
2420
IC Score
1
2
Iteration Step

Backwise Step Best Model Regression Summary

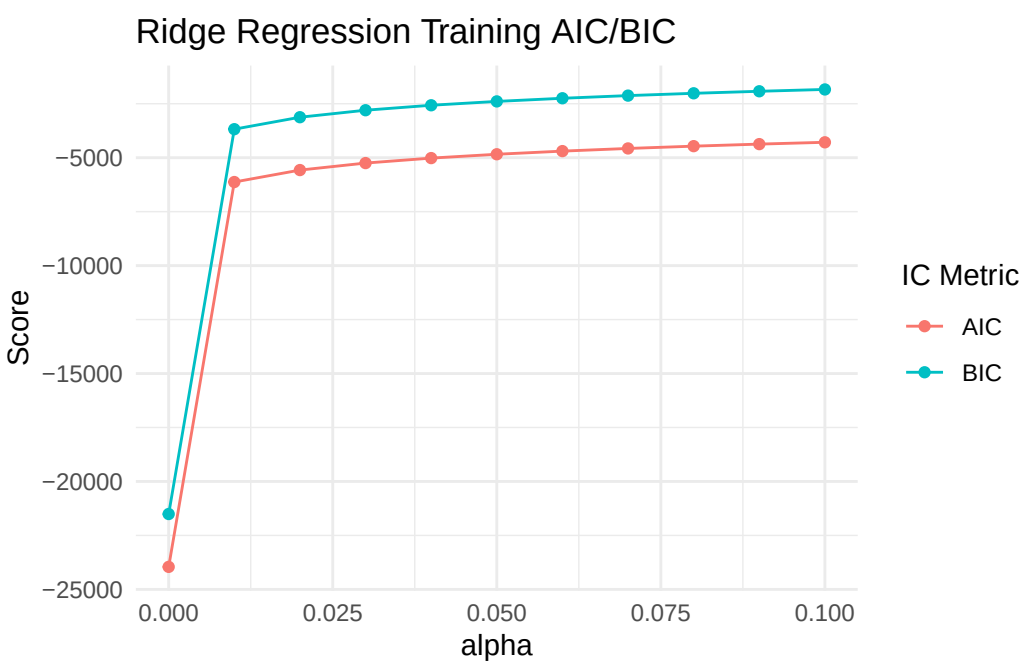
	medv
crim	-0.13***
	(0.04)
zn	0.06***
	(0.02)
chas1	1.79*
	(0.95)
nox	-20.42***
	(4.05)
rm	3.70***
	(0.48)
dis	-1.65***
	(0.22)
rad	0.28***
	(0.07)
tax	-0.01***
	(0.004)
ptratio	-1.01***
	(0.15)
lstat	-0.49***
	(0.05)
Constant	44.31***
	(5.61)
N	400
R2	0.74
Adjusted R2	0.73
Residual Std. Error	4.83 (df = 389)
F Statistic	109.63*** (df = 10; 389)

```
==========================================================
Notes:                ***Significant at the 1 percent level.
                       **Significant at the 5 percent level.
                       *Significant at the 10 percent level.
```

Backward Step Regression Results:

1. The step model reached the lowest AIC (~2408) and BIC(~2456) after 3 iterations with 10 out of the 12 features retained.
2. Retained Features: **crime, zn, chas, nox, rm, dis, rad, tax, ptratio and lstat** with the **age** and **indus** variables removed**.** All of these are still statistically significant at least up to the 10% level.
3. All regressors also maintain their sign with very little changes to their maginal effects with respect to the response variable.
4. R square (~74%) and the F statistic (significant at 1%) are still comparable and in the same region as the baseline model

Ridge Regression



Best lambda val=0 for AIC consideration

```
13 x 1 sparse Matrix of class "dgCMatrix"
               lambda.1se
(Intercept) 25.4992007212
crim        -0.0704068185
zn           0.0248067620
indus       -0.0746079757
chas         2.1120254227
nox         -5.8133328458
rm           3.4638733478
age         -0.0113346933
dis         -0.4766797473
rad         -0.0006444531
tax         -0.0037944410
ptratio     -0.6941725792
lstat       -0.3124781100
```

```
13 x 1 sparse Matrix of class "dgCMatrix"
              lambda.1se
(Intercept) 20.443661358
crim        -0.023273616
zn           .
indus        .
chas         0.797518744
nox         -3.825130295
rm           4.325835327
age          .
dis         -0.155948750
rad          .
tax         -0.002066481
ptratio     -0.850391951
lstat       -0.463540792
```

```
[1] 16.64220 16.64976 16.96179 19.77408  0.00000
```

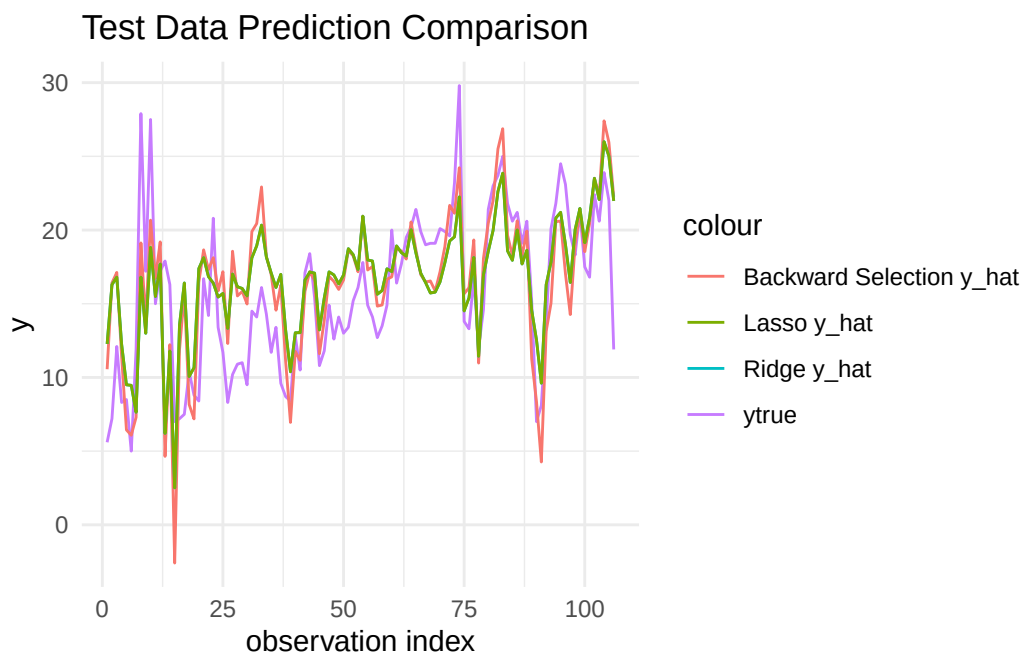
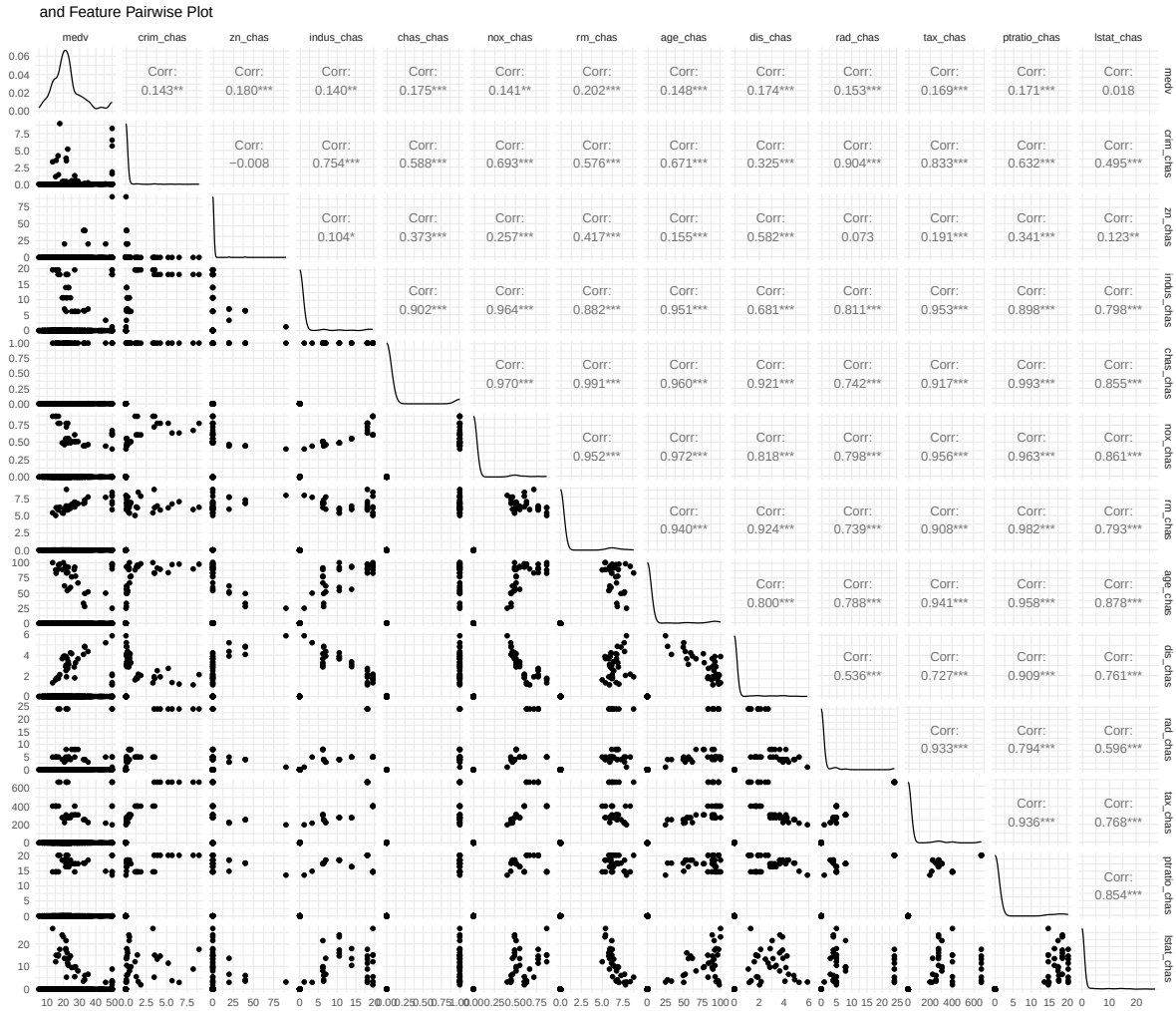


Table 4: Model MSE Metrics

metric	all_features	backwise_step	ridge	lasso
MSE	16.6422	16.64976	16.96179	16.96179

*best model= all features model (yikes!)

https://statisticsbyjim.com/regression/interaction-effects/



Median House Values, medv, shows the highest correlation with the rm_chas interaction variable. Look at and this correlation also happens to be statistically significant at 5%.

```
Call:
lm(formula = medv ~ ., data = int_tr_d)

Residuals:
    Min      1Q  Median      3Q     Max
-13.175  -2.938  -0.783   1.849  27.677

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  44.526464   5.733305   7.766 7.36e-14 ***
```

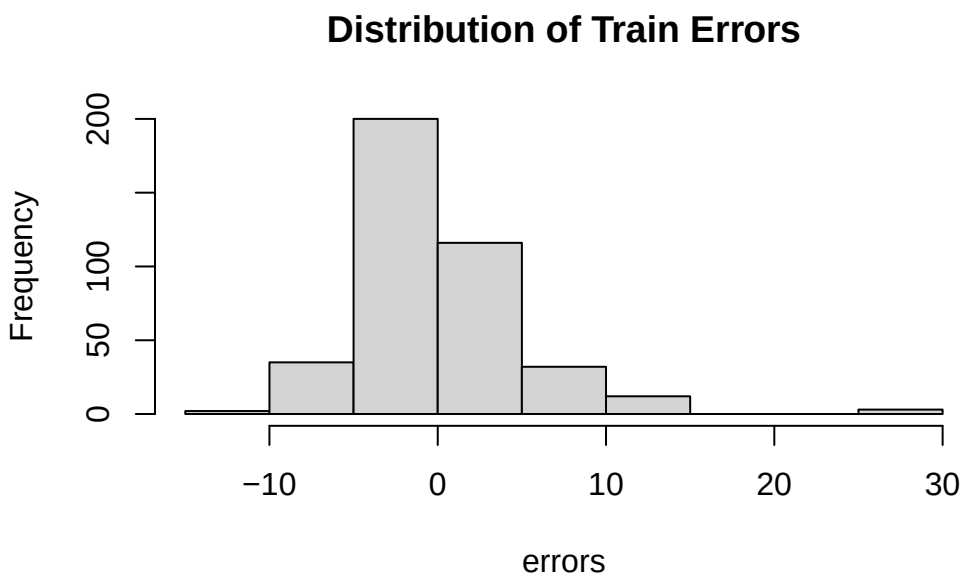
```
crim         -0.127376   0.039281  -3.243 0.001287 **
zn            0.055996   0.015879   3.526 0.000472 ***
indus         0.032937   0.077219   0.427 0.669953
chas          1.184828   7.117119   0.166 0.867870
nox         -20.577159   4.420878  -4.655 4.47e-06 ***
rm            3.734252   0.510945   7.309 1.57e-12 ***
age          -0.005220   0.015165  -0.344 0.730850
dis          -1.650745   0.232964  -7.086 6.61e-12 ***
rad           0.286979   0.079776   3.597 0.000363 ***
tax          -0.014810   0.004696  -3.154 0.001739 **
ptratio      -1.014128   0.147635  -6.869 2.60e-11 ***
lstat        -0.480899   0.057936  -8.301 1.76e-15 ***
rm_chas       0.088848   1.076285   0.083 0.934251
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.851 on 386 degrees of freedom
Multiple R-squared:  0.7383,    Adjusted R-squared:  0.7295
F-statistic: 83.77 on 13 and 386 DF,  p-value: < 2.2e-16
```

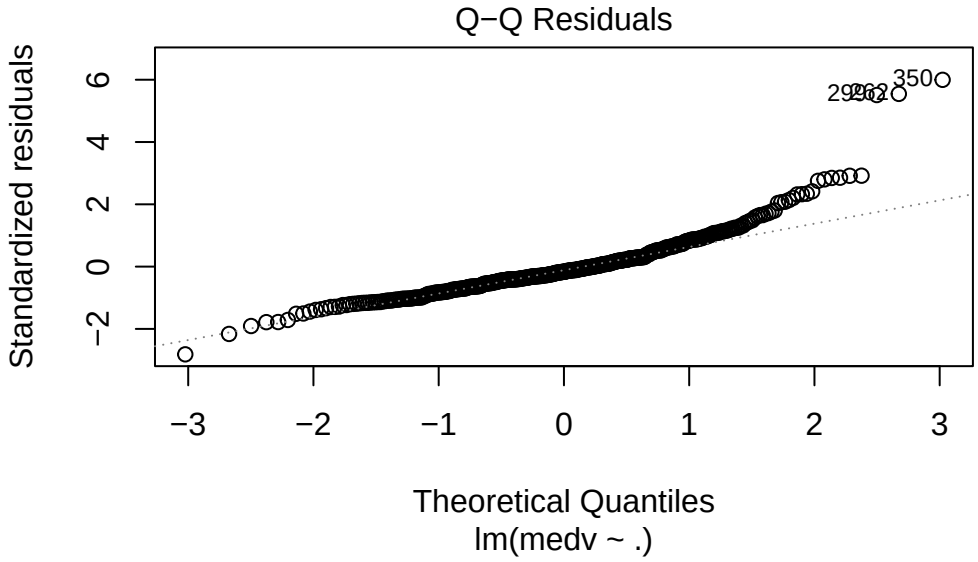
rm_chas did not turn out to be significant- but I'm not sure if this is as a result of the data imbalance (lot more chas=0 than 1) ?

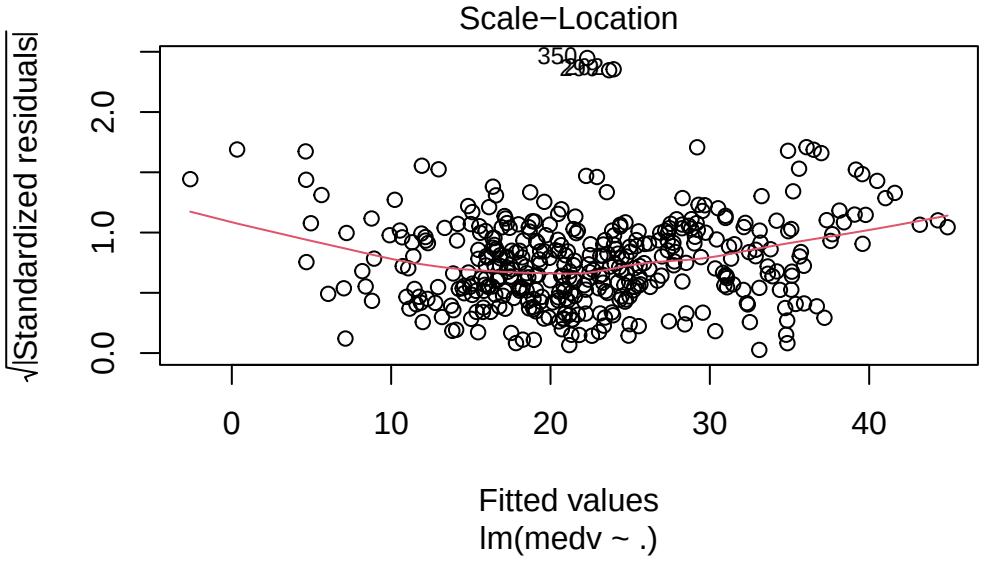
```
[1] 16.61995
```

```
[1] 16.6422
```

Residual Diagnostics




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
lm(medv ~ .)
350


Scale−Location
√|Standardized residuals|
Fitted values
lm(medv ~ .)
350